

The first $\log n$: the two-dimensional standard integral

$$Z_2(n) \sim \frac{1}{4} S_{1/2}(a) n^{-1/2} \log n$$

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Abstract

The first genuinely multivariate result of the grammar arc and the paper's first $\log n$: for the chart integral with $d = 2$, $k = (1, 1)$, $h = (0, 0)$, $\xi \equiv a$, $\eta \equiv 1$,

$$Z_2(n) = \int_0^b \int_0^b e^{-\beta n u^2 v^2 + \beta \sqrt{n} u v a} dv du \sim \frac{S_{1/2}(a)}{4} n^{-1/2} \log n \quad (n \rightarrow \infty),$$

with the explicit bound $|Z_2(n) - \frac{A}{2} n^{-1/2} \log n| \leq (2A|\log b| + E + M)n^{-1/2}$ for $\sqrt{n}b^2 \geq 1$, $A = S_{1/2}(a)/2$. The candidate exponent $\mu = \frac{1}{2}$ appears with multiplicity $m = 2$, exactly as the state density $-(\log \tau)/(4\sqrt{\tau})$ of $\tau = u^2 v^2$ predicts. No two-dimensional Fubini theorem is used. Zero sorry/axiom.

1 Setting

With the one-dimensional theory complete, the next frontier was the $\log n$ powers of `thm:TaylorTree`, which arise only for $d \geq 2$ from repeated poles of the state density's Mellin transform. Rather than build the Mellin/state-density machinery, a GPT-6 Astra consult (`tide-log/gpt6_log2d_v1.md`) suggested a route through two one-dimensional substitutions and a squeeze: with $f(s) = e^{-\beta s^2 + \beta a s}$, $F(x) = \int_0^x f$, and the logarithmic primitive $H(L) = \int_0^L F(x)/x dx$, the inner integral is $F(\sqrt{n}ub)/(\sqrt{n}u)$, the outer then gives $Z_2(n) = n^{-1/2}H(\sqrt{n}b^2)$, and $H(L) = A \log L + O(1)$ uniformly. The logarithm is the integrated $1/u$ between the saturation scale $u \sim 1/(b\sqrt{n})$ and the fixed chart endpoint.

2 The things formalised

`QuadraticKernel.lean`: `quadKernel`, `quadPrimitive`, `quadMass` ($= S_{1/2}(a)/2$), `quadMoment` ($= S_1(a)/2$), `logPrimitive`; Gaussian domination; $0 \leq F(x) \leq Ex$; $0 \leq A - F(x) \leq M/x$; $\int_1^L dx/x^2 = 1 - 1/L$; and the squeeze

```
theorem logPrimitive_bounds ( a L : ) (h : 0 < ) (hL : 1 ≤ L) :
  quadMass a * Real.log L - quadMoment a logPrimitive a L
  logPrimitive a L quadMass a * Real.log L + Real.exp ( - a ^ 2 / 2 )
```

`LogTwoD.lean`: `twoD_inner`, `twoDIntegral_eq` ($Z_2 = n^{-1/2}H(\sqrt{n}b^2)$), `twoDIntegral_sub_le` (the explicit bound), `isLittle0_inv_sqrt_log`, and

```
theorem twoDIntegral_isEquivalent ( a b : ) (h : 0 < ) (hb : 0 < b) :
  (fun n : => twoDIntegral a b n) ~[atTop]
  (fun n : => fluctuation (1 / 2) a / 4 * (n ^ (-(1 / 2 : )) * Real.log n)
```

3 Proof strategy

Everything is one-dimensional. Mass and moment are the unit-22 kernel identity at $n = 1$, $k = 1$, $h = 0, 1$. The primitive is an interval integral (continuity from `intervalIntegral.continuous_primitive`); $F(x) \leq Ex$ by `integral_mono_on` against a constant; $A - F(x) = \int_x^\infty f \leq M/x$ by comparing $xf(s) \leq sf(s)$. For H , split $(0, L]$ at 1: on $(0, 1]$ the integrand $F(x)/x \leq E$; on $(1, L]$ write $F/x = A/x - (A - F)/x$, integrate A/x exactly to $A \log L$ (`integral_one_div_of_pos`) and bound $(A - F)/x \leq M/x^2$ with $\int_1^L dx/x^2 \leq 1$ (`integral_zpow`). The two substitutions are `intervalIntegral.integral_comp_mul_left`; the outer one uses the everywhere-valid identity $F(cu)/u = c \cdot F(cu)/(cu)$ (both sides vanish at $u = 0$ under Lean's division convention). Finally $\log(\sqrt{n}b^2) = \frac{1}{2} \log n + 2 \log b$, and $n^{-1/2} = o(n^{-1/2} \log n)$ via $1/\log n \rightarrow 0$.

4 Roadblocks and resolutions

- open `intervalIntegral` together with `MeasureTheory` makes `integral_const_mul`, `integral_sub` ambiguous; keep the `intervalIntegral.` prefix explicit.
- `integrableOn_const` now has `autoParam` side conditions that `aesop` could not discharge for `volume (Ioc 0 L)` ; pass `measure_Ioc_lt_top.ne` explicitly.
- `intervalIntegrable_inv/integralIntegrable_zpow` are not the installed names; integrability of $1/x$ and $1/x^2$ on $(1, L]$ was obtained by bounded domination instead (robust, no name dependence). `Set.Ioc_disjoint_Ioc_same` likewise absent — prove disjointness directly.
- `simp only [...]` at `h` on the unit-22 identity mangled the integrand before it could be matched; rewrite the integrand to the kernel first (`setIntegral_congr_fun`), then simplify the constants with `norm_num`.
- Pointwise goals of `IntegrableOn.congr_fun` arrive un-beta-reduced: `beta_reduce` before `rw`. `Integrable.const_mul/.sub` return a bare `Integrable` (an `And`) — type-ascribe as `IntegrableOn` before `.congr_fun/.mono_set`.
- `rw [...]` at `*` silently left the squeeze hypothesis in its old form; rewrite the log identity into that hypothesis explicitly so that `linarith` sees identical atoms.
- In the last step, `Real.rpow_neg` found no $(n^y)^{-1}$ because the goal had a division; rewrite `Real.rpow_neg` forwards on the other side and let `ring` finish.

5 What was Mathlib, what was new

Mathlib: `intervalIntegral.integral_comp_mul_left`, `integral_of_le`, `continuous_primitive`, `integral_mono_on`, `integral_one_div_of_pos`, `integral_zpow`, `Real.log_sqrt`, `tendsto_log_atTop`, `Tendsto.inv_tendsto_atTop`, `IsEquivalent.congr_right`. New: the whole logarithmic-primitive theory and the first formal $\log n$ asymptotic for a standard integral. The leading constant was checked numerically two ways ($S_{1/2}(a)/4 = 0.512920$ against the erf closed form) and the squeeze verified at $n = 10^2, \dots, 10^8$ before any Lean was written.

6 Lessons

When an expansion is expected to carry a logarithm, look for a one-dimensional reduction in which the logarithm is literally $\int dx/x$ over a growing range; the uniform $O(1)$ squeeze of the logarithmic primitive is much cheaper than any meromorphic-continuation argument, and it needs no Fubini theorem because the inner integral is a primitive.

7 Follow-ups

The exact next-order term ($-B n^{-1/2}$ with the log-moment $B = \int f \log s$), the product-measure form of Z_2 via Fubini, general (k_1, k_2) and (h_1, h_2) in two dimensions (multiplicity 2 iff the two candidate exponents coincide), and the general- d multiplicity structure.